

$$V(x, y) = \frac{1}{2} k (x^2 + y^2) + \lambda W(x, y) \quad (I) W = (x^2 + y^2)^2 \quad (II) W = x^2 y^2$$

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - V(x, y)$$

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} k (x^2 + y^2) - \lambda W(x, y)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = m \ddot{x} \quad \frac{\partial \mathcal{L}}{\partial x} = -kx - \lambda \frac{\partial W}{\partial x}$$

$$m \ddot{x} + kx + \lambda \frac{\partial W}{\partial x} = 0$$

$$\ddot{x} + \Omega^2 x = -\frac{\lambda}{m} \frac{\partial W}{\partial x}$$

$$\ddot{y} + \Omega^2 y = -\frac{\lambda}{m} \frac{\partial W}{\partial y}$$

(I)

$$\frac{\partial W}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2)^2 = 2(x^2 + y^2) \cdot 2x = 4x(x^2 + y^2)$$

$$\frac{\partial W}{\partial y} = 2(x^2 + y^2) \cdot 2y = 4y(x^2 + y^2)$$

$$\ddot{x} + \Omega^2 x = -\frac{4\lambda}{m} x (x^2 + y^2)$$

$$\ddot{y} + \Omega^2 y = -\frac{4\lambda}{m} y (x^2 + y^2)$$

$$\ddot{r} + \Omega^2 r = -\frac{4\lambda}{m} r r^2$$

(II)

$$\ddot{x} + \Omega^2 x = -\frac{2\lambda}{m} x y^2$$

$$\ddot{y} + \Omega^2 y = -\frac{2\lambda}{m} x^2 y$$